Math 2551 Worksheet: Curvature and Normals

1. Find the unit tangent vector, unit normal vector, and curvature of the curve $\mathbf{r}(t) = \langle \sqrt{2}t, e^t, e^{-t} \rangle$, $t \in \mathbb{R}$.

2. Find $\mathbf{T}, \mathbf{N}$ and κ for the space curve $\mathbf{r}(t) = (\cos(t) + t\sin(t))\mathbf{i} + (\sin(t) - t\cos(t))\mathbf{j} + 3\mathbf{k}$ with $t \geq 0$.

3. Compute $\mathbf{T}$ and $\mathbf{N}$ for the curve $\mathbf{r}(t) = \langle t, (1/3)t^3 \rangle, t \in \mathbb{R}$ for $t \neq 0$.

   Does $\mathbf{N}$ exist at $t = 0$? Graph the curve and explain what is happening to $\mathbf{N}$ as t passes from negative to positive values.

4. Before doing any computations, where do you think that the curvature of the parabola $y = x^2$ is greatest?

   Compute its curvature and find the point with greatest curvature.